

Research Paper

FDEM-flow3D: A 3D hydro-mechanical coupled model considering the pore seepage of rock matrix for simulating three-dimensional hydraulic fracturing

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ARTICLE INFO

Article history:

Received 25 February 2016

Received in revised form 15 July 2016

Accepted 17 August 2016

Available online 29 August 2016

Keywords:

Finite discrete element method (FEMDEM)

Hydraulic fracturing

Hydro-mechanical coupling

Pore seepage

Calibration aperture

Fluid-driven fractures

ABSTRACT

A hydro-mechanical coupled model that can simultaneously consider the pore seepage of a rock matrix and the fracture seepage is proposed to simulate three-dimensional hydraulic fracturing. This model appropriately takes into account the fluid leak-off into the surrounding rock matrix from the fracture. Several examples are given to validate the seepage algorithms and the coupled model. The results suggest that this model can solve problems involving pore seepage and fracture seepage through simple pure fracture seepage. Moreover, it can reproduce the fluid pressure distribution and the crack initiation and propagation and consider the fluid loss during hydraulic fracturing.

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1. Introduction

Hydraulic fracturing is widely used in the energy industry to improve the exploitation of oil, gas and geothermal energy. This technique is a typical rock fracture phenomenon driven by fluid that involves not only hydro-mechanical coupling but also crack initiation and propagation and interactions between natural fractures and hydraulic fractures [1]. In this process, in addition to the fluid flow within the fracture (i.e., fracture seepage), there is also fluid that leaks into the rock matrix (i.e., pore seepage). Therefore, the simulation of hydraulic fracturing is believed to be one of the most challenging problems in the field of geomechanics. Conventional research has focused on the hydro-mechanical coupling in the existing complex fracture network without regard for the crack initiation and propagation and without considering fracturing fluid leakage into the fractured rock [2–8] (i.e., the pore seepage in the rock matrix is not considered).

Within the analytical models of hydraulic fracturing, there is the main PKN model [9,10], the KGD model [11,12], and the radial or penny-shaped model [13], as well as some quasi three-dimensional (P3D) and three-dimensional models (PL3D) [14]. However, these models are mainly applied to simple fracture mor-

phologies, and most of them consider only the fluid flow in the fracture but not the fluid leak-off into the reservoir from the fracture, i.e., they may not also consider pore seepage in the rock matrix during hydraulic fracturing. Given the limitations of the analytical models, many scholars have studied hydraulic fracturing using numerical simulation technology. The research in this field can be divided into two categories, the methods based on continuum mechanics and the approaches based on discontinuous mechanics.

Within the methods based on continuum mechanics, Pan et al. [15] combined discontinuous cellular automata and the multi-phase flow software “Tough” to simulate the interaction of multiple fractures in the injection process of CO₂. Zhang et al. [16] studied natural reservoir fracture driven by fluid using the discontinuous displacement method. They assumed that closed natural fractures allow fluid inflow, whereby the fluid leaking into the natural fracture from newly generated fractures can be considered. However, the rock matrix is still impervious, and therefore the fluid leak-off into the rock matrix from the fractures is not considered. Fu et al. [17] proposed a two-dimensional explicit hydro-mechanical coupled model to simulate hydraulic fracturing with an arbitrary fracture network. However, the fluid flow occurs in only the fractures, without regard to fluid leak-off into the surrounding rock matrix. Hu et al. [18] built a non-linear fluid-solid interaction finite element model to simulate hydraulic fracturing using the smeared crack model. In this model, the fracture seepage

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and pore seepage are both considered. Bazant and Ohtsubo [19] analyzed hydraulic fracturing using the finite element method. However, the problem with the finite element method is that the mesh must be in accordance with the boundary of the fractures, which will require constant re-meshing to trace the crack propagation during the calculation [20,21]. The extended finite element method (XFEM), though, can avoid re-meshing [22]. For example, Salimzadeh and Khalili [23] proposed a fully coupled three-phase model for simulating hydraulic fracturing in porous media, where XFEM is used to handle discontinuity and a cohesive crack model is used as a fracturing criterion. Using this model, they investigated the leakage of fracturing fluid into the medium around the crack tip. However, XFEM requires the use of a very fine mesh in the solution domain when the crack path is unknown in advance, which will lead to a huge amount of computation [18]. In addition, XFEM cannot easily simulate complex crack propagation and intersection.

Based on the concept of the amended Terzaghi effective stress and Biot consolidation theory, Yang and Tang et al. [24,25] developed a rock fracture analysis system (F-RFPA2D) considering damage and seepage coupling. Although F-RFPA2D is based on the finite element method, it can simulate crack initiation and propagation, permeability evolution and the mechanism of seepage-stress coupling during hydraulic fracturing. This analysis system was later developed into the three-dimensional case by combination with parallel technology [26,27].

Within the approaches based on discontinuous mechanics, besides the particle discrete element method, methods such as discontinuous deformation analysis (DDA) and the discrete element method (UDEC, 3DEC) usually cannot directly simulate block fracture. However, these methods can simulate crack initiation and propagation by introducing virtual joints between sub-blocks. Thus, in combination with a hydro-mechanical coupling model, they can be used to simulate hydraulic fracturing. For example, Ben [28,29] and Jiao et al. [30–32] introduced a virtual joint and a hydro-mechanical coupled model to DDA to simulate hydraulic fracturing. Morgan and Aral [33], based on the work of Ben et al. [28,29], developed a fully implicit algorithm for modeling hydraulic fracturing. They used a finite volume fracture network model to describe the compressible fluid flow in fractures. Nasehi and Mortazavi [34] studied the effect of the in situ stress, the intact rock strength parameters (c , ϕ), the elastic modulus and the discontinuities on hydraulic fracturing by introducing a virtual joint into UDEC. Using the same technology in 3DEC, Hamidi and Mortazavi [35] simulated the crack initiation and propagation driven by fluid in a typical reservoir. It should be noted that these methods do not consider fluid leak-off into the surrounding rock matrix from a fracture, although they can simulate the crack initiation and propagation, the fluid flow in a fracture, and the interaction between a fluid and solid. In other words, these methods consider fracture seepage but do not consider pore seepage in the rock matrix, i.e., the permeability of the rock matrix has been ignored.

The particle discrete element method (PFC) proposed by Cundall [36] is another approach based on discontinuous mechanics, which uses a series of balls bonded together to characterize the continuum and simulates the crack initiation and propagation by breaking the bonds between balls [37–39]. The method is a very promising technique for simulating hydraulic fracturing. For example, Shimizu et al. [40] developed a particle discrete element model coupling flow to study the effect of the fluid viscosity and particle size distribution on hydraulic fracturing. The model can consider both seepage flow and pore seepage. Damjanac et al. [41,42] proposed a discrete lattice model with hydro-mechanical coupling to simulate hydraulic fracturing in unconventional reservoirs that overcomes the main drawbacks of the traditional methods (including DEM) and exhibits a higher computational efficiency than that of PFC.

To simulate the transition from continua to discontinua, Munjiza [43,44] proposed the combined finite-discrete element method (FEMDEM) in 1995. This method can simulate the whole process of material deformation and fracture, incorporating the advantages of the discrete element method while retaining the concepts of stress and strain from the finite element method. Therefore, it is an excellent method to simulate rock fracture. However, this method does not well consider fluid-solid interactions, and no seepage-stress coupling calculation module is offered. For example, Grasselli et al. [45] studied the effect of discontinuities on crack propagation driven by fluid pressure using the two-dimensional FEMDEM, but they just took the fluid pressure as a uniformly distributed load on the fissure without considering fluid flow in the fissure as well as the effect of the fissure's opening or closure on fluid flow. For this reason, many researchers combined this method with other fluid flow simulation software to indirectly implement seepage-stress coupling and to study the stress-dependent permeability in fractured rock [46–48].

Clearly, the recent studies of the FEMDEM described above cannot directly solve rock fracture driven by fluid, but can do so only in combination with existing fluid flow simulation software to indirectly implement seepage-stress coupling. Alternatively, they may not consider the fluid flow at all and instead take the fluid pressure in the injection hole as a load uniformly distributed on the crack. Yan et al. [49] proposed a hydro-mechanical coupled model that can simulate hydraulic fracturing with an arbitrarily complex fracture network in the two-dimensional case but does not yet consider fluid leak-off into the surrounding rock matrix from the fracture. However, the leak-off of reservoir fracturing fluid occurs even in a low-permeability reservoir, and in a high-permeability reservoir, the fluid leak-off phenomenon is more violent. Therefore, it is necessary to modify the model to consider pore seepage. Moreover, because the actual hydraulic fracturing is a three-dimensional problem, we should extend the model to three dimensions to simulate hydraulic fracturing more realistically to guide fracturing construction. Consequently, in this paper we propose a hydro-mechanical coupled model (namely, FDEM-flow3D) considering pore seepage to simulate three-dimensional hydraulic fracturing.

This paper is organized as follows. First, the fundamental principles of mechanical rupture calculation and the system of equations in the FEMDEM are briefly introduced. The fracture model of a joint element with six nodes in the three-dimensional FEMDEM is highlighted, which is the key to using the FEMDEM to simulate rock fracture. Then, we construct a three-dimensional hydro-mechanical coupled model on the basis of the FEMDEM that can consider pore seepage and the fluid tangential viscous force. In addition, we propose a calibration method for determining the initial hydraulic aperture of the joint elements to characterize the permeability of the rock matrix. Finally, three examples are given, in which the first is used to verify the correctness of FDEM-flow3D for handling a pure pore seepage problem, the second verifies the hydro-mechanical model, and the last is used to validate the correctness of FDEM-flow3D in the simulation of hydraulic fracturing and to verify that the model is able to simultaneously capture fracture seepage and pore seepage in a rock matrix during hydraulic fracturing.

2. Fundamentals of mechanical rupture calculation and the fracture model of joint elements with six nodes in the 3D FEMDEM

The basic idea of the three-dimensional FEMDEM is that a continuum is divided into a finite element mesh with tetrahedral elements and six-node joint elements with bonding effects are

inserted in the common surface between adjacent tetrahedral elements as shown in Fig. 1. The method uses the finite element method to obtain the stress and strain of constant-strain tetrahedral elements, while the discrete element method is used to address the contact between tetrahedral elements. Thus, the FEM-DEM can simulate the sliding and rotating of the block. In addition, the method can simulate crack initiation, propagation and intersection in a continuum by breaking joint elements. Before a joint element is broken, the deformation of the continuum can be simulated by joint elements with bonding effects and deformable tetrahedral elements. Here, we briefly introduce the kinetic equation of the FEMDEM, with an emphasis on the fracture model of joint elements with six nodes in the three-dimensional FEMDEM.

2.1. System equations of the FEMDEM

In the FEMDEM, the load and element mass are treated as nodes to build the dynamic equation of a system according to Newton's second law

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) \quad (1)$$

where $\mathbf{M}$ and $\mathbf{C}$ are the mass matrix and damping matrix of all nodes, respectively, and $\mathbf{F}(\mathbf{x})$ represents the total nodal force vector which includes the node force vector $\mathbf{F}_c(\mathbf{x})$ assigned by the contact force between tetrahedral elements, the node force vector $\mathbf{F}_d$ induced by the deformation of tetrahedral elements, the node force vector $\mathbf{F}_j(\mathbf{x})$ caused by the deformation of joint elements, and the node force vector $\mathbf{F}_e(\mathbf{x})$ assigned by the external load.

The introduction of the damping matrix $\mathbf{C}$ consumes the kinetic energy of the system, by which we use dynamic relaxation to solve the static problem. The damping matrix is given by

$$\mathbf{C} = \mu \mathbf{I} \quad (2)$$

where μ is the damping coefficient, and $\mathbf{I}$ is a unit matrix. For a mass spring system with single degree of freedom, the critical damping coefficient is given by [44]

$$\mu = 2h\sqrt{\rho E} \quad (3)$$

where h is the edge length of an element, ρ is the element density, and E is the elastic modulus of the element. The system kinetic energy will be consumed at the fastest rate if the critical damping is used.

The solution of Eq. (1) uses the same method as the DEM, namely, that the kinetic equation of the nodes is solved by the Euler method. The coordinates and velocity of the nodes of the tetrahedral elements at each time step are updated by the formula

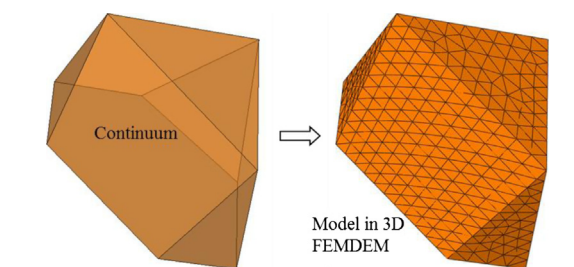
$$\begin{aligned} v_i^{(t+\Delta t)} &= v_i^{(t)} + \sum F_i^{(t)} \frac{\Delta t}{m_n} \\ x_i^{(t+\Delta t)} &= x_i^{(t)} + v_i^{(t)} \Delta t \end{aligned} \quad (4)$$

where $F_i^{(t)}$ represents a component of the total nodal force vector, Δt is the time step, and m_n is the node mass, which is equal to one quarter of the tetrahedral element.

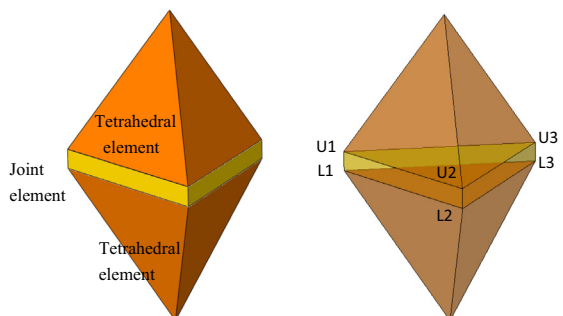
2.2. Fracture model of joint elements in the three-dimensional FEMDEM

In the 3D FEMDEM, a six-node joint element with bonding effects is inserted in the common surface between two adjacent tetrahedral elements. The combined single and smeared crack model [50,51] is used to characterize the stress-opening relationship or the failure criterion of the joint element. The crack initiation and propagation in the continuum is implemented by broken joint elements. As shown in Fig. 2, because the joint element shares nodes with two adjacent tetrahedral elements, crack propagation occurs along the boundary of the tetrahedral elements.

As shown in Fig. 2, the coordinate vector of the top three nodes of a joint element is $\mathbf{U}_i$ ($i = 1, 2, 3$), whereas the coordinate vector of the three nodes at the bottom surface of the joint element is $\mathbf{L}_i$ ($i = 1, 2, 3$). Thus, the three nodes at the middle section of the joint element can be given by

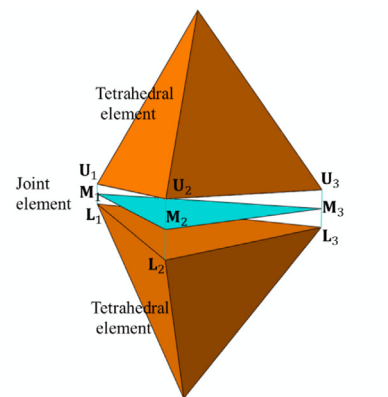


(a) Continuum representation in the 3-dimensional FEMDEM

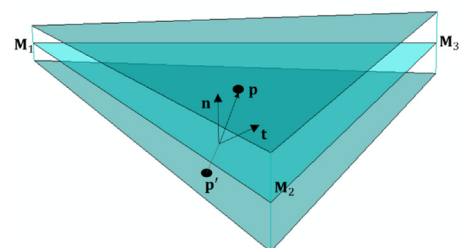


(b) Connection of two tetrahedral solid elements with a joint element

Fig. 1. Mesh configuration in the three-dimensional FEMDEM.



(a) Connection of a joint element with two tetrahedral elements



(b) Opening amount calculation of a joint element with six nodes

Fig. 2. Diagram for the opening amount calculation of a joint element.

$$\mathbf{M}_i = \frac{\mathbf{U}_i + \mathbf{L}_i}{2} \quad (5)$$

Initially, the top surface overlaps with the bottom surface, but any two overlapping points at the top and bottom surfaces will no longer coincide when the joint element is open. Thus, the open vector of the joint element at this point can be expressed as

$$\delta = \mathbf{p} - \mathbf{p}' \quad (6)$$

where $\mathbf{p}$ and $\mathbf{p}'$ are the coordinate vectors of two points in two tetrahedral elements, which initially coincide.

The open vector can be decomposed into

$$\delta = \delta_n \mathbf{n} + \delta_t \mathbf{t} \quad (7)$$

where δ_n and δ_t are the normal component and tangential component of the open vector, respectively, namely, the normal opening amount and the tangential slippage amount; $\mathbf{n}$ is the normal unit vector of the middle section of the joint element; and $\mathbf{t}$ is the unit vector along the projection direction of the open vector in the middle section. The two unit vectors can be expressed as

$$\mathbf{n} = \frac{(\mathbf{M}_2 - \mathbf{M}_1) \times (\mathbf{M}_3 - \mathbf{M}_1)}{|(\mathbf{M}_2 - \mathbf{M}_1) \times (\mathbf{M}_3 - \mathbf{M}_1)|} \quad (8)$$

$$\mathbf{t} = \delta - (\delta \cdot \mathbf{n})\mathbf{n}$$

Thus, the normal opening amount and tangential slippage amount can be given by

$$\begin{aligned} \delta_n &= \delta \cdot \mathbf{n} \\ \delta_t &= \delta \cdot \mathbf{t} \end{aligned} \quad (9)$$

Similarly, the stress vector can be decomposed into

$$\mathbf{T} = \sigma \mathbf{n} + \tau \mathbf{t} \quad (10)$$

We describe the relationships between the normal stress component and the normal opening amount and between the tangential stress component and the tangential slippage amount in the following section.

In theory, the normal opening amount of a point is equal to zero (i.e., $\delta_n = \delta_{np} = 0$) when the normal stress is equal to or less than the tensile strength (i.e., $\sigma = f_t$). In other words, there should be no opening of the joint elements before the normal stress reaches the tensile strength. However, the opening amount of a joint element in the FEMDEM will never be zero because two adjacent tetrahedral elements do not share the same nodes. To ensure that the opening amount between two tetrahedral elements is small enough, a penalty function method is used, which adds a normal spring to apply a normal constraint force on the two tetrahedral elements.

When $0 < \delta_n \leq \delta_{np}$, the normal stress of a joint element can be expressed as

$$\sigma = [2\delta_n/\delta_{np} - (\delta_n/\delta_{np})^2]f_t \quad (11)$$

where $\delta_{np} = 2hf_t/p_n$ is the normal opening amount when the normal stress is equal to the tensile strength f_t , h is the edge length of the joint element, and p_n is the normal penalty parameter. When p_n tends to infinity, $\delta_{np} = \lim_{p_n \rightarrow +\infty} 2hf_t/p_n = 0$. That is to say, if the normal penalty parameter is large enough, the normal opening amount of the joint elements will tend to zero before the normal stress reaches the tensile strength.

When $\delta_{np} < \delta_n \leq \delta_{nc}$ (δ_{nc} is the critical normal opening amount when the normal stress decreases to zero), the material will exhibit strain softening (i.e., the normal stress gradually decreases) as the normal opening amount increases. Thus, the normal stress can be expressed as

$$\sigma = zf_t \quad (12)$$

where z is a strain softening expression that is obtained by fitting the stress-strain curves of a material

$$z = \left[1 - \frac{a+b-1}{a+b} e^{d\{a+cb/[(a+b)(1-a-b)]\}} \right] \cdot [a(1-d) + b(1-d)^c] \quad (13)$$

where a, b, c are obtained by fitting the experimental curves of the material under tension, and d is a dimensionless quantity to characterize the crack opening, which is given by

$$d = \begin{cases} \sqrt{\left(\frac{\delta_n - \delta_{np}}{\delta_{nc}}\right)^2 + \left(\frac{|\delta_t| - \delta_{tp}}{\delta_{tc}}\right)^2}, & \delta_{nc} > \delta_n > \delta_{np} \text{ and } \delta_{ts} > |\delta_t| > \delta_{tp} \\ \frac{\delta_n - \delta_{np}}{\delta_{nc}}, & \delta_{nc} > \delta_n > \delta_{np} \text{ and } |\delta_t| \leq \delta_{tp} \\ \frac{\delta_t - \delta_{tp}}{\delta_{tc}}, & \delta_n \leq \delta_{np} \text{ and } \delta_{ts} > |\delta_t| > \delta_{tp} \\ 1, & \text{if } \delta_n \geq \delta_{nc} \text{ or } |\delta_t| \geq \delta_{tc} \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

where δ_t is the tangential slippage, δ_{tp} is the tangential slippage amount when the tangential stress is equal to the shear strength f_s , and δ_{tc} is the critical tangential slippage amount when the tangential stress decreases to zero.

If $d \geq 1$, then the joint element will break, and the two tetrahedral elements that were connected by the joint element will be completely disconnected.

When $\delta_n \leq 0$, the joint element is in a compression state. To make the embedded amount between two tetrahedral elements small enough, we again use the penalty function method, which adds a normal spring to apply a normal repulsion between the two tetrahedral elements. The normal stress can be expressed as

$$\sigma = 2 \frac{\delta_n}{\delta_{np}} f_t \quad (15)$$

Thus, the complete relationship between the normal stress and the normal opening amount is obtained by

$$\sigma = \begin{cases} 2 \frac{\delta_n}{\delta_{np}} f_t, & \delta_n < 0 \\ \left[\frac{2\delta_n}{\delta_{np}} - \left(\frac{\delta_n}{\delta_{np}}\right)^2 \right] f_t, & 0 \leq \delta_n \leq \delta_{np} \\ zf_t, & \delta_n > \delta_{np} \end{cases} \quad (16)$$

Similar to the above, the relationship between the tangential stress and the tangential slippage can be expressed as

$$\tau = \begin{cases} zf_s, & \sigma > 0 \text{ and } |\delta_t| > \delta_{tp} \\ \left[\frac{2|\delta_t|}{\delta_{tp}} - \left(\frac{|\delta_t|}{\delta_{tp}}\right)^2 \right] zf_s, & \sigma > 0 \text{ and } |\delta_t| \leq \delta_{tp} \\ zf_s - \sigma \tan \varphi, & \sigma \leq 0 \text{ and } |\delta_t| > \delta_{tp} \end{cases} \quad (17)$$

where φ is the friction angle of the joint element.

3. Basic principles and assumptions of FDEM-flow3D

As shown in Fig. 1, the tetrahedral elements and joint elements in the FEMDEM have a unique topological connection. We assume that the tetrahedral elements are impervious, and the fluid flow occurs in only joint elements. Although the permeability of the rock matrix is small compared with that of the discontinuities, the rock matrix still has a small permeability (The fluid flow in the rock matrix is the pore seepage). For this reason, we assumed that all of the joint elements allow fluid inflow or outflow, which is different from the assumption in the literature [49]. If a joint element is not broken, it still allows fluid inflow or outflow to characterize the permeability of the rock matrix. However, when a joint element is broken, the opening amount of the joint element will

increase rapidly under the effect of the fluid pressure due to the loss of the bonding force. Then, the permeability of the broken joint element also increases dramatically and is larger than that of the unbroken joint elements to characterize the permeability of the discontinuities in the rock mass.

Because the rock mass is composed of discontinuities and the rock matrix that is cut by the discontinuities, the permeability of the rock mass also consists of two parts: the permeability of the rock matrix (corresponding to pore seepage) and the permeability of the discontinuities in the rock mass (corresponding to fracture seepage). Thus, the FDEM-flow3D model considering pore seepage is based on the following assumptions: (1) all the tetrahedral elements are impermeable; (2) all the joint elements allow fluid inflow and outflow; (3) the permeability of the rock matrix is characterized by that of the unbroken joint elements; and (4) the permeability of the discontinuities (including newly generated cracks and natural fractures) in the rock mass is represented by that of the broken joint elements.

4. FDEM-flow3D model considering pore seepage (i.e., rock matrix permeability)

Based on the above assumptions, we transform the problems mixing pore seepage and fracture seepage into pure fracture seepage problems. Thus, FDEM-flow3D still uses the fracture seepage to describe the fluid flow in rock mass. The hydro-mechanical coupled model includes the following process: fluid flow in a joint element will apply fluid pressure on the two tetrahedral elements connected by the joint element. Then, the two tetrahedral elements will separate or close under the effect of the fluid pressure and external load. This process corresponds to the open or closing of the joint element, while the joint element opening or closing will affect the fluid flow in the joint element and change the fluid pressure in turn.

As shown in Fig. 3, the fluid flow in the joint element will apply pressure on the two tetrahedral elements connected by the joint element (see Fig. 3 (a)). Then, the two tetrahedral elements will separate, and the opening amount of the joint element increases (see Fig. 3 (b)). According to the cubic law, the flow rate per unit width is proportional to the cube of the hydraulic aperture, which directly depends on the opening amount of the joint element. Thus,

the joint element opening will affect the flow rate and fluid pressure in turn (see Fig. 3 (b)).

Assume that in FDEM-flow3D, the tetrahedral elements and joint elements constitute a topological structure, as shown in Fig. 4(a). The topological structure can be decomposed into joint elements, as shown in Fig. 4(b), and tetrahedral elements, as shown in Fig. 4(c). The joint elements constitute a two-dimensional planar passage for fluid flow, while the tetrahedral elements are impermeable. Therefore, the fluid flow in FDEM-flow3D is essentially a two-dimensional flow problem in space. Because the fluid flow occurs in only the zone shown in Fig. 4(b), the tetrahedral elements shown in Fig. 4(c) are excluded from the seepage calculation. The common nodes of adjacent triangular surfaces in Fig. 4(b) constitute the intersection of the seepage calculation, such as nodes 1–7 in Fig. 4(b). Because the fluid pressure distribution inside a triangular surface may not be uniform, we use the linear interpolation of the fluid pressure at three nodes to represent the distribution of the fluid pressure in the entire triangular surface. Any triangular surface (i.e., a joint element) constitutes a two-dimensional planar flow domain. As shown in Fig. 4(b), the triangular surface $\Delta 123$ constitutes a two-dimensional planar flow domain. Here, we introduce how to calculate the fluid flow on an arbitrary triangular surface based on the topological structure, as shown in Fig. 4(b).

Take node 1 in Fig. 4(b) as an example. The triangular surfaces connected at node 1 are $\Delta 123$, $\Delta 134$, $\Delta 145$, $\Delta 125$, $\Delta 126$, $\Delta 136$, $\Delta 146$, $\Delta 156$, $\Delta 127$, $\Delta 137$, $\Delta 147$ and $\Delta 157$. Take triangular surface $\Delta 123$ as an example to describe the solution steps of the fluid flow within a triangular surface. Assume the total head at nodes 1, 2 and 3 is ϕ_1 , ϕ_2 and ϕ_3 , respectively. The coordinates of the three nodes in the local coordinate system (x -axis and y -axis located within the plane of the triangular surface and z -axis perpendicular to the triangular surface) are $(x_k, y_k, 0)$ (where k is the node number, $k = 1, 2, 3$). Supposing that the total head within the triangular surface obeys a linear distribution, then the head gradient at any point within the triangular surface is constant and can be expressed as

$$\frac{\partial \phi}{\partial x_i} = \frac{1}{A} \int_A \frac{\partial \phi}{\partial x_i} dA \quad (18)$$

where A is the area of the triangular surface, ϕ is the total head of a point within the triangular surface, and x_i ($i = 1, 2$) is the coordinate components along the i direction in the local coordinate system.

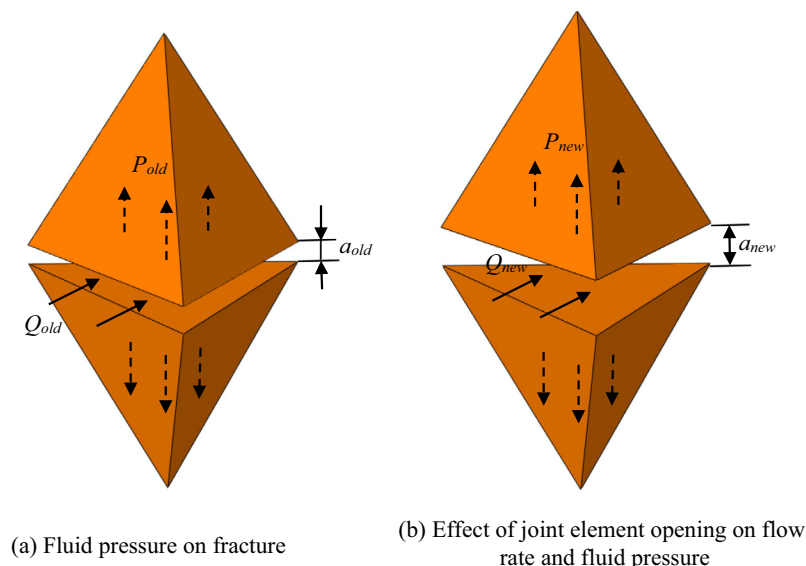


Fig. 3. Fundamentals of hydro-mechanical coupling.

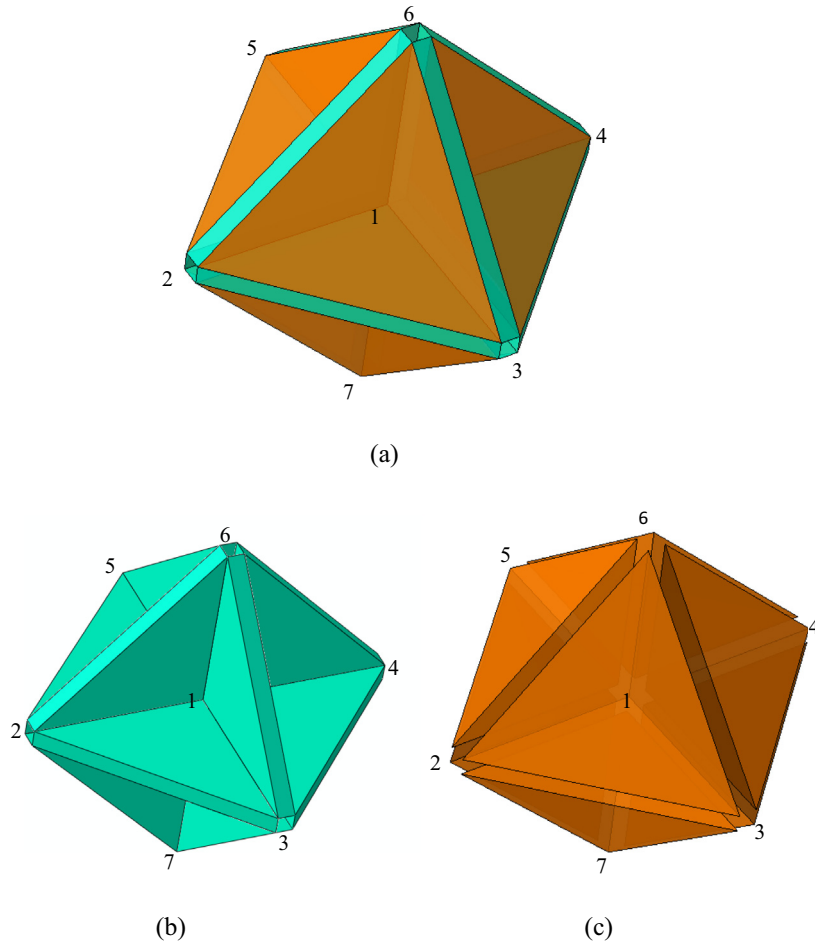


Fig. 4. Fluid flow in mesh combining joint elements and tetrahedral elements.

According to the divergence theorem, Eq. (18) can be written as

$$\frac{\partial \phi}{\partial x_i} = \frac{1}{A} \int_s \phi n_i ds = \frac{1}{A} \sum_{m=1}^3 \bar{\phi}^m \in_{ij} \Delta x_j^m \quad (19)$$

where n_i is the outward normal unit vector, $\bar{\phi}^m$ is the average fluid pressure at edge m , Δx_j^m is the difference in coordinate component between two vertices of edge m , and $\in_{ij}$ is the two-dimensional permutation tensor

$$\in = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

According to the cubic law, the flow rate along the i direction can be expressed as

$$q_i = -\frac{a^3 \rho_f g}{12\mu} \frac{\partial \phi}{\partial x_i} \quad (20)$$

where q_i is the flow rate in the i direction, a is the hydraulic aperture of the joint elements, ρ_f is the fluid density, μ is the fluid dynamic viscosity coefficient, $\phi = z + p/\rho_f g$ is the total head, z is the elevation, g is the gravitational acceleration, and p is the fluid pressure.

Obviously, the hydraulic aperture of a joint element is not a constant. Its relationship with the average normal opening amount $\bar{\delta}_n$ of a joint element is shown in Fig. 5.

Thus, the hydraulic aperture of a joint element can be expressed as

$$a = a_0 + \bar{\delta}_n \quad (21)$$

where a_0 represents the initial hydraulic aperture of the joint element, and $\bar{\delta}_n$ is the average normal opening amount of a joint element, with a positive value denoting opening. Because the hydraulic aperture cannot be less than or equal to zero in the seepage calculation, we set a minimum hydraulic aperture $a_{\min}$. If the hydraulic aperture a according to Eq. (21) is smaller than $a_{\min}$, a is set to $a_{\min}$. To describe the permeability dropping when a joint element closes, $a_{\min}$ can be set to a small value. The average normal opening amount of a joint element may take the average value of the normal opening amount at the three endpoints, i.e., $\bar{\delta}_n = (\delta_{n1} + \delta_{n2} + \delta_{n3})/3$, where δ_{n1} , δ_{n2} , and δ_{n3} are the normal open-

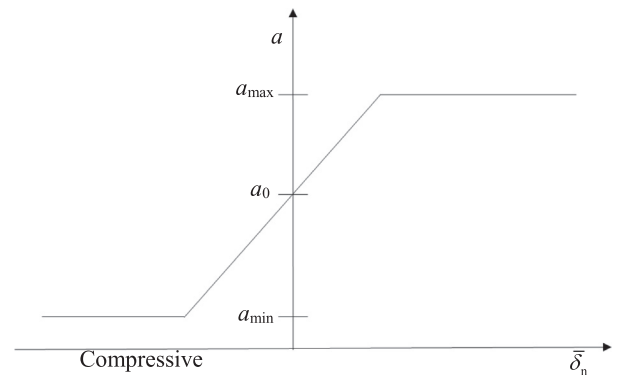


Fig. 5. Relationship of the hydraulic aperture with the average normal opening amount of a joint element.

ing amount of the three endpoints at the top and bottom surfaces of the joint element, as shown in Fig. 6.

Because the fluid flow in Eq. (20) is the microscopic flow of a triangular surface while the fluid flow in the whole numerical rock model is composed of many two-dimensional triangular surfaces, the macroscopic permeability of the numerical rock model cannot be directly obtained by Eq. (20). However, we can determine the initial hydraulic aperture a_0 of the joint elements by simulating a permeability test for the numerical rock model. If the macroscopic permeability of the numerical rock model at a given initial hydraulic aperture is consistent with that of the actual rock, this initial hydraulic aperture is just what we need. This means that we can update the permeability and porosity of the numerical rock model by adjusting the initial hydraulic aperture a_0 of the joint elements.

It should be noted that when the fluid pressure at the three nodes of a triangular surface is zero, the flow rate according to Eq. (20) is not zero because of gravity. That is unreasonable because the flow rate should decrease as the saturation decreases, so that fluid should not flow out from the zone with zero saturation. For this purpose, Eq. (20) is multiplied by a coefficient f_s , which is a function of the average saturation $\bar{s}$

$$f_s = \bar{s}^2(3 - 2\bar{s}) \quad (22)$$

where $\bar{s} = (s_1 + s_2 + s_3)/3$, with s_1 , s_2 and s_3 as the saturation values of nodes 1, 2 and 3, respectively. This function has the characteristics of $f_s = 0$ if $\bar{s} = 0$ and $f_s = 1$ if $\bar{s} = 1$. This means that the flow rate inside the triangular surface is zero if the average saturation is zero, whereas the flow rate inside the triangular surface is not reduced if the node is fully saturated. Moreover, the derivative of the function at $\bar{s} = 0, 1$ is zero, so it is physically reasonable. Therefore, if Eq. (20) is multiplied by the coefficient f_s , the flow will not occur in a zone with zero saturation. Thus, the final flow rate along the i direction can be expressed as

$$q_i = -\frac{a^3 \rho_f g}{12\mu} \frac{\partial \phi}{\partial x_i} f_s \quad (23)$$

Then, substituting Eq. (19) into (23), we can obtain the flow rate along the x and y directions.

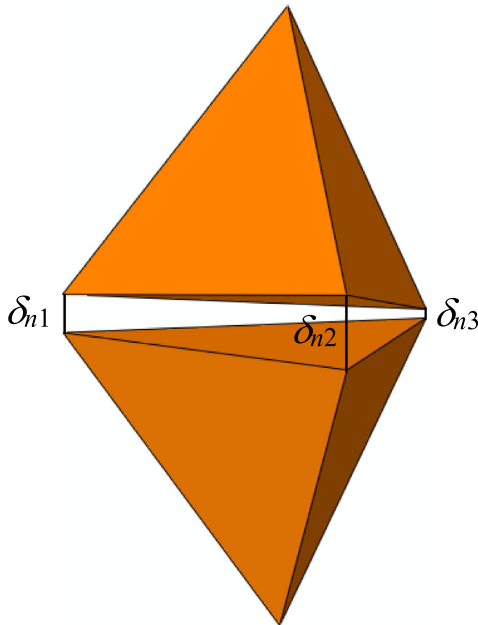


Fig. 6. Normal opening amounts of the endpoints of a joint element.

According to Fig. 7, the flow into node 1 per unit time can be calculated using the formula

$$Q = Q_x + Q_y = A_y q_x + A_x q_y \quad (24)$$

where A_x and A_y are the differences in the coordinate components between the midpoints of the two edges through node 1.

Thus, we have obtained the flow $Q_{\Delta 123}$ into node 1 from the triangular surface $\Delta 123$. Similarly, we can obtain the flow into node 1 from the other triangular surfaces that directly connect to node 1. Finally, the total flow into node 1 per unit time can be expressed as

$$Q_{\text{total}} = Q_{\Delta 123} + Q_{\Delta 127} + Q_{\Delta 176} + Q_{\Delta 156} + Q_{\Delta 145} + Q_{\Delta 134} \quad (25)$$

Then, the current pressure of node 1 can be obtained according to the formula

$$p = p_0 + K_w Q \frac{\Delta t}{V} - K_w \frac{\Delta V}{V_m} \quad (26)$$

where p_0 is the fluid pressure of node 1 at the previous time step, K_w is the bulk modulus of the fluid, Q is the total flow of node 1 (inflow is positive, outflow is negative), Δt is the time step, $\Delta V = V - V_0$, and $V_m = (V + V_0)/2$, where V and V_0 are the node volumes at the current time step and the previous time step.

If the node pressure according to Eq. (26) is negative, then the node pressure is set to zero, indicating that the fluid flowing into node 1 is not enough to fill the equivalent volume of the node. Thus, the saturation of node 1 will decrease and will be updated according to the formula

$$s = s_0 + Q \frac{\Delta t}{V} - \frac{\Delta V}{V_m} \quad (27)$$

where s_0 is the saturation of node 1 at the previous time step. If the node saturation $s < 0$ at the current time step, the fluid pressure of the node will remain unchanged. However, the saturation will be set to one if $s > 1$. Through this strategy, the fluid mass conservation is ensured. If the fluid pressure according to Eq. (26) is greater than zero, the change in the node volume will alter the node pressure. On the contrary, the change in the node volume will alter the node saturation if the node pressure according to Eq. (27) is less than zero.

To ensure the numerical stability of the seepage algorithm, the time step should not exceed the threshold

$$\Delta t_f = \min \left[\frac{V}{K_w \sum_i \frac{a_i^3}{12\mu}} \right] \quad (28)$$

where a_i is the hydraulic aperture of the i th joint element connecting to the node.

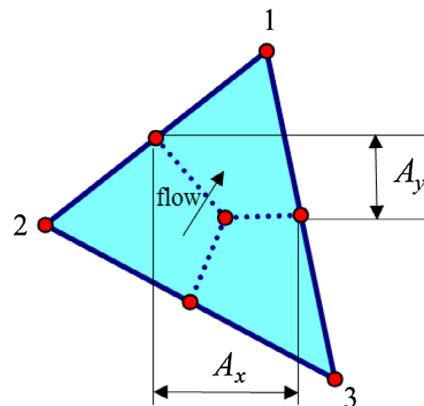


Fig. 7. Fluid flow into node 1.

5. Calibration of the initial hydraulic aperture of the joint elements to characterize the permeability of the rock matrix

Because the macroscopic permeability of a numerical rock model depends on the initial hydraulic aperture of the joint elements, we can determine an appropriate initial hydraulic aperture to characterize the permeability of the rock matrix. The appropriate initial hydraulic aperture a_0 should make the macroscopic permeability of the numerical rock model obtained by the permeability test simulation consistent with that of the actual rock sample.

As shown in Fig. 8, a rock numerical model with length L , height H and width W is used for the permeability test simulation. Initially, the numerical rock model is totally saturated. The fluid pressure at the left boundary surface is maintained at P_{in} , while that at the right boundary surface is P_{out} . The front, rear, bottom, and top boundary surfaces are impermeable. Thus, fluid will flow through all the joint elements in the model under the pressure difference. When the inflow Q_{in} equals the outflow Q_{out} ($Q = Q_{in} = Q_{out}$), the simulation of the permeability test is terminated because the fluid flow within the numerical rock model has reached the steady state at this time.

It is assumed that the entire numerical rock model is an isotropic medium. According to Darcy's law, the flow per unit time at steady state can be calculated by

$$Q = \frac{k \cdot W \cdot H}{\mu} \frac{(P_{in} - P_{out})}{L} \quad (29)$$

where μ and k are the fluid viscosity coefficient and the macroscopic permeability of the numerical rock model, respectively. Thus, the macroscopic permeability of the numerical rock model can be calculated by the formula

$$k = \frac{Q\mu L}{W \cdot H(P_{in} - P_{out})} \quad (30)$$

Choose an initial hydraulic aperture a_0 , and perform the simulation of the permeability test as shown in Fig. 8. Then, according to Eq. (30), the macroscopic permeability k of the numerical rock model has been obtained. If the macroscopic permeability k is exactly equal to the permeability of the actual rock sample, the appropriate initial hydraulic aperture is obtained. In this case, the permeability of the joint elements with the initial hydraulic aperture a_0 exactly characterizes the permeability of the rock matrix. However, if the macroscopic permeability does not equal that of the actual rock sample, we need to adjust the initial hydraulic aperture and continue the permeability test simulation until the macroscopic permeability of the numerical rock model is consistent with that of the actual rock sample. This is the calibration method by which we determine a reasonable value for the initial hydraulic aperture to take into account the permeability of the rock matrix.

Once cracks are generated in the rock (i.e., joint elements are broken), the hydraulic apertures of the broken joint elements will

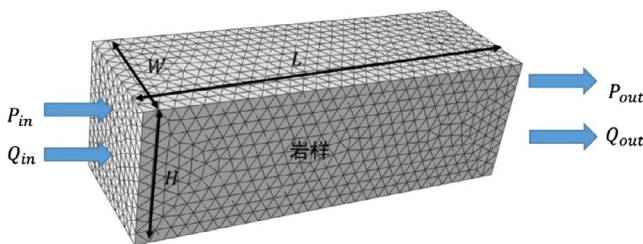


Fig. 8. Numerical rock model for the simulation of the permeability test.

significantly increase. According to Eq. (20), the permeability of the broken joint elements will also significantly increase, whereby we can characterize the permeability of the discontinuities in the rock mass.

It can be observed that the benefits of FDEM-flow3D are that we can use simple pure fracture seepage to simultaneously characterize the pore seepage and fracture seepage.

6. Fluid pressure and fluid tangential viscous force

The fluid pressure of the nodes can be obtained according to the preceding seepage calculation. Thus, the fluid pressure distribution on the boundary surfaces of the tetrahedral elements can be obtained by linear interpolation according to the fluid pressures of three nodes in one boundary surface. Then, a linearly distributed fluid pressure is applied on the triangular surfaces of two tetrahedral elements connected by a joint element along the inward normal direction. As shown in Fig. 9(a), assume that the fluid pressures at nodes 1, 2 and 3 are p_1 , p_2 and p_3 , respectively. The total normal fluid pressure that is applied on two tetrahedral elements connected by a joint element is given by

$$f_n = \frac{p_1 + p_2 + p_3}{3} A \quad (31)$$

In addition to the fluid pressure, the FDEM-flow3D model also takes into account the tangential viscous force of the fluid, which has been implemented in our program.

It is assumed that the fluid flow within the joint elements obeys Poiseuille flow, i.e., parallel plate laminar flow. The flow direction in a joint element is shown in Fig. 9(b), and we select a cross-section along the flow direction. The flow velocity distribution in the cross-section is shown in Fig. 9(c), where the distances from both the bottom and top of the joint element to the middle section are $r = a/2$. Because the fluid flow is assumed to be parallel plate laminar flow, the fluid flow velocity perpendicular to the flow direction is zero. Thus, the fluid flow velocity between the two parallel plates composed of two triangular surfaces is given by

$$\begin{aligned} v_x &= \frac{1}{2\mu} \left[\left(\frac{a}{2} \right)^2 - r^2 \right] \frac{\partial p}{\partial x} \\ v_y &= \frac{1}{2\mu} \left[\left(\frac{a}{2} \right)^2 - r^2 \right] \frac{\partial p}{\partial y} \end{aligned} \quad (32)$$

According to Newton's internal friction law, the shear stress distribution on the effective interface can be obtained by the following equation, i.e., at an arbitrary point from which the distance to the middle section is r , the shear stress is given by

$$\begin{aligned} \tau_x &= -\mu \frac{dv}{dr} = r \frac{\partial p}{\partial x} \\ \tau_y &= -\mu \frac{dv}{dr} = r \frac{\partial p}{\partial y} \end{aligned} \quad (33)$$

In particular, in the parallel plate ($r = a/2$), the shear stress is given by

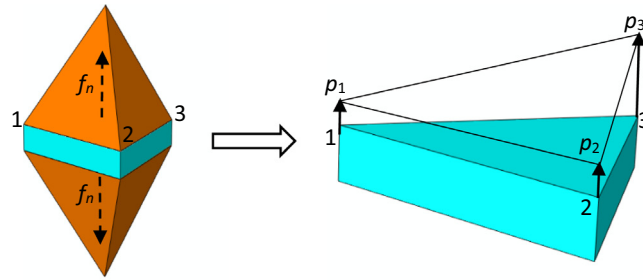
$$\begin{aligned} \tau_{fx} &= \frac{a}{2} \frac{\partial p}{\partial x} \\ \tau_{fy} &= \frac{a}{2} \frac{\partial p}{\partial y} \end{aligned} \quad (34)$$

The total tangential viscous force applied on two triangular surfaces of the tetrahedral elements is given by

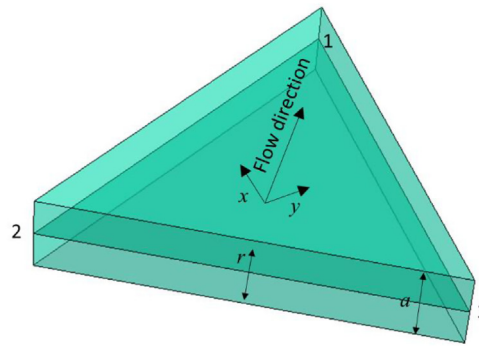
$$\begin{aligned} f_{cx} &= A \tau_{fx} = \frac{a}{2} A \frac{\partial p}{\partial x} \\ f_{cy} &= A \tau_{fy} = \frac{a}{2} A \frac{\partial p}{\partial y} \end{aligned} \quad (35)$$

where A is the area of a triangular surface, i.e., the area of the middle section of a joint element in FDEM-flow3D.

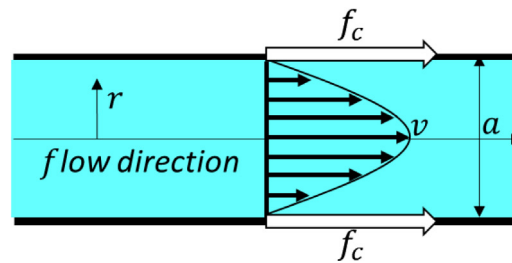
Thus, the fluid pressure and tangential viscous force on the solid tetrahedral elements are all determined through the above algorithm. Then, the fluid effect force is applied on the two triangular surfaces of the tetrahedral elements connected by a joint element



(a) Fluid pressure distribution



(b) 3D schematic of the direction of fluid flow in a joint element



(c) Fluid flow velocity and tangential viscous force distribution within the cross-section along the flow direction

Fig. 9. Fluid flow and forces in a joint element.

as a linearly distributed surface load. Just like the contact forces and external load, the fluid effect force is also assigned to each node of the tetrahedral elements, and thus the application of the fluid force is complete.

7. Calculation procedure

The preceding sections mainly focus on how to perform the seepage calculation and how to apply the fluid effect force. In this section, we introduce the FDEM-flow3D model from the view of combining the mechanical calculation and seepage calculation, as shown in Fig. 10. The calculation procedures of the FDEM-flow3D model considering pore seepage are as follows. First, the nodal force induced by the deformation of the tetrahedral elements and joint elements is obtained according to the nodal displacement

at the previous time step. Second, the contact detection is performed, and then the contact force between adjacent tetrahedral elements is calculated. The contact force is also assigned to the nodes of the tetrahedral elements. Third, the seepage calculation is performed, and the nodal fluid pressure and tangential viscous force of the joint elements are updated. After that, the fluid effect force is applied on the triangular surfaces of the tetrahedral elements connected by joint elements as a linearly distributed surface load and is assigned to the nodes of the tetrahedral elements. Finally, the total nodal force is obtained, and the velocity and coordinates of all nodes are updated according to Newton's second law. Thus, a complete time step including the seepage and mechanical calculations is finished. Following the procedures above, the seepage and mechanical coupled calculation is continued until the time step number reaches the set value.

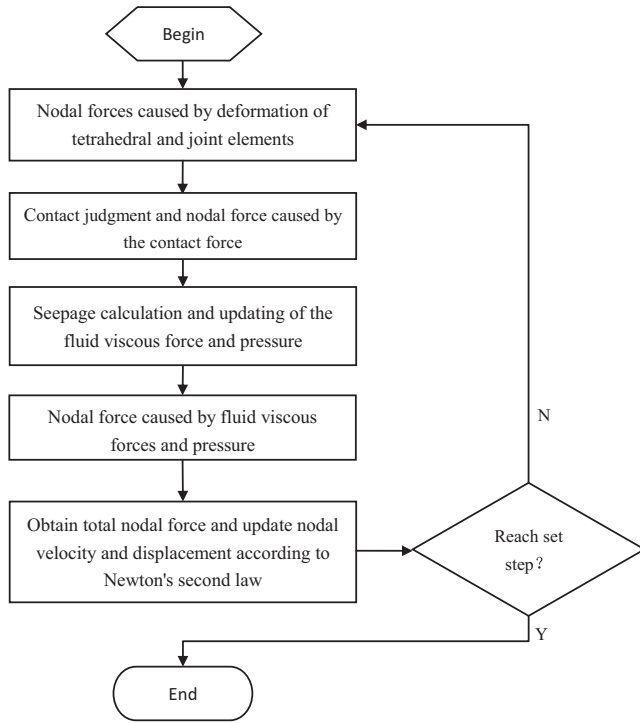


Fig. 10. Calculation procedure of FDEM-flow3D model considering pore seepage.

8. Examples

8.1. Unidirectional flow in two different permeability formations

As shown in Fig. 11, the permeability of segment L_1 is k_1 , while at segment L_2 the permeability is k_2 . The fluid pressure on the left boundary surface is constant at p_1 , and that on the right boundary surface is denoted as p_2 . The fluid pressure at the interface is denoted as p_m . Assuming that the seepage in the formations obeys Darcy's law, then the fluid flow through segment L_1 can be given by

$$q_1 = \frac{k_1 F}{\mu} \cdot \frac{p_1 - p_m}{L_1} \quad (36)$$

The fluid flow through segment L_2 is given by

$$q_2 = \frac{k_2 F}{\mu} \cdot \frac{p_m - p_2}{L_2} \quad (37)$$

According to the continuity of steady state flow $q_1 = q_2 = q$, the fluid pressure at the interface is given by

$$p_m = \frac{k_1 L_2 p_1 + k_2 L_1 p_2}{k_1 L_2 + k_2 L_1} \quad (38)$$

Thus, in the interval of $0 < x < L_1$, the fluid pressure at a point is equal to

$$p = p_1 - \frac{q\mu}{k_2 F} x = p_1 - \frac{p_1 - p_2}{k_1 \left(\frac{L_1}{k_1} + \frac{L_2}{k_2} \right)} x \quad (39)$$

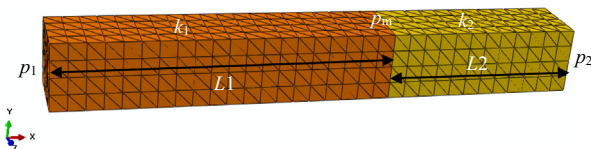


Fig. 11. Unidirectional flow in two different permeability formations.

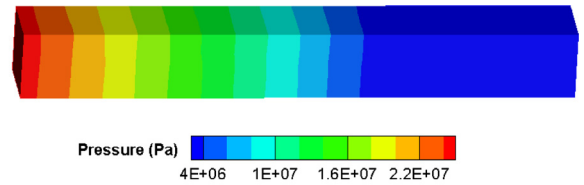
Similarly, in the interval of $L_1 < x < L_1 + L_2$, the fluid pressure at a point is equal to

$$\begin{aligned} p &= p_m - \frac{q\mu}{k_2 F} (x - L_1) \\ &= p_1 - \left(\frac{1}{k_1} - \frac{1}{k_2} \right) \frac{p_1 - p_2}{k_1 \left(\frac{L_1}{k_1} + \frac{L_2}{k_2} \right)} L_1 - \frac{p_1 - p_2}{k_1 \left(\frac{L_1}{k_1} + \frac{L_2}{k_2} \right)} x \end{aligned} \quad (40)$$

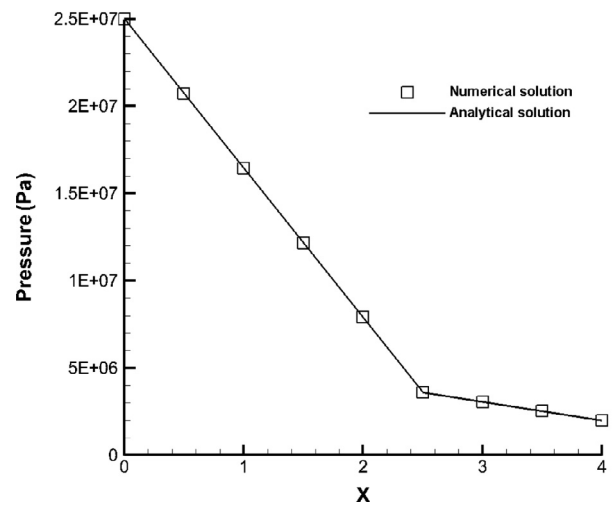
The flow through the entire formation per unit time can be obtained by

$$q = \frac{p_1 - p_2}{\frac{\mu}{F} \left(\frac{L_1}{k_1} + \frac{L_2}{k_2} \right)} \quad (41)$$

We simulate the example using the FDEM-flow3D model to verify the correctness of the model in dealing with a pure pore seepage problem. As shown in Fig. 11, the model was meshed into 3079 tetrahedral elements and 8502 joint elements with an average size of 0.25 m. The front, rear, top and bottom boundary surfaces are impermeable, while the pressure at the left boundary surface is fixed at 25 MPa and that at the right boundary surface is fixed at 2 MPa. To characterize the different permeabilities of the left and right segments, we take different hydraulic apertures for segments L_1 and L_2 . The hydraulic aperture of the joint elements in segment L_1 is 3×10^{-5} m, while that in segment L_2 is 1.5×10^{-5} m. The macroscopic permeabilities of the two segments are $k_1 = 9.804 \times 10^{-15}$ m² and $k_2 = 7.84 \times 10^{-14}$ m² according to the calibration method in Section 5. The fluid bulk modulus K_w is 2.2×10^9 Pa, and the fluid viscosity coefficient μ is 1 mPa s. In this case, all the nodes are fixed regardless of the mechanical calculation.



(a) Fluid pressure distribution in different permeability formations



(b) Numerical solution and analytical solution of the fluid pressure distribution along the x-axis direction

Fig. 12. Comparison of the numerical solution and the analytical solution of the fluid pressure distribution.

tion, as only the seepage calculation is needed. The time step is 1×10^{-5} s. The final fluid pressure obtained by the FDEM-flow3D model as shown in Fig. 12.

As shown in Fig. 12(a), the fluid pressure decreases from left to right. It is clear that the lengths of the part with the same color are

long in the right segment L_2 but are short in the left segment L_1 , which means that the fluid pressure in the left segment L_1 drops more quickly than that in the right segment L_2 , i.e., the permeability in the left segment L_1 is smaller than that in the right segment L_2 , as shown in Fig. 12(b). The fluid pressure along the x -axis dis-

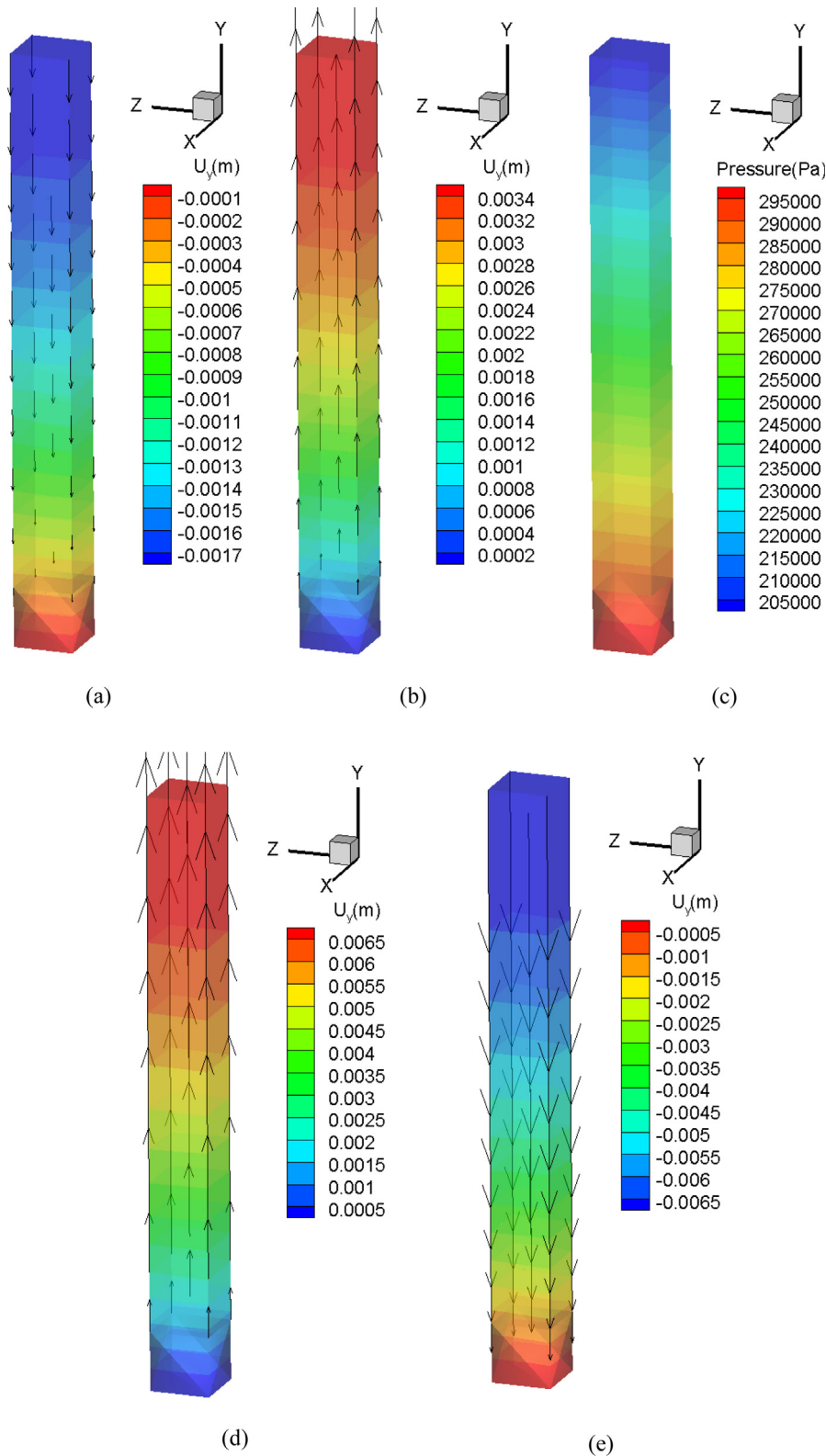


Fig. 13. Simulation result using the coupled hydro-mechanical model.

plays a piecewise linear distribution, with the turning point of the fluid pressure curve along the x -axis located at the interface ($x = 2.5$ m). The fluid pressure gradient (corresponding to the slope of the line) in the left segment L_1 is larger than that in the right segment L_1 . The numerical solution of the fluid pressure along the x -axis is in good agreement with the analytical solution.

The flow through the entire formation per unit time according to the analytical formula (20) is $2.098 \times 10^{-5} \text{ m}^3/\text{s}$, whereas that obtained by FDEM-flow3D is $2.099 \times 10^{-5} \text{ m}^3/\text{s}$. Clearly, the fluid flow obtained by the model is almost identical to that obtained by the analytical solution.

8.2. A poroelastic example illustrating the solid weight, buoyancy and seepage forces

A cuboid has height $H = 10$ m and mechanical boundary conditions that the base and sides of the model are fixed along the normal direction. The cuboid is elastic with a Young's modulus of $E = 81.818 \text{ MPa}$, a Poisson's ratio of $\nu = 0.364$, and a density of $\rho_d = 500 \text{ kg/m}^3$. The bulk modulus of water is 2.2 GPa , the density of water is $\rho_w = 1000 \text{ kg/m}^3$, and the gravitational acceleration g is set to 10 m/s^2 . 孔隙率 $n=0.5$

Initially, the water table is at the bottom of the cuboid, and the cuboid is in equilibrium under gravity. The heave of the layer when the water level is raised and the heave or settlement under a vertical head gradient are studied. Because this simple problem has an analytical solution [52], we use it to verify the coupled hydro-mechanical model in this paper.

In stage (1), the water table is at the bottom of the cuboid. Find the displacement of the model at the top under gravity, the analytical solution of which is given by

$$u_y = -\frac{\rho_d g H^2 (1 + \nu)(1 - 2\nu)}{2E(1 - \nu)} \quad (42)$$

The calculated value from the coupled hydro-mechanical model is $-1.780\text{e}-3 \text{ m}$, which matches the analytical value ($-1.786\text{e}-3 \text{ m}$) at this stage. The displacement distribution is shown in Fig. 13(a).

In stage (2), continue from stage (1) and reset the displacements to zero; then, the water table is raised to the top of the cuboid, and the saturation is initialized to 1 and fixed at the top. Find the displacement of the cuboid at the top under buoyancy, the analytical solution of which is given by

$$u_y = \frac{\rho_w g H^2 (1 + \nu)(1 - 2\nu)}{2E(1 - \nu)} * (1 - n) \quad (43)$$

The calculated value from the coupled hydro-mechanical model is $+3.588\text{e}-3 \text{ m}$, which matches the analytical value ($+3.572\text{e}-3 \text{ m}$) at this stage. The displacement distribution is shown in Fig. 13(b).

In stage (3), continue from stage (2) and again reset the displacements to zero and raise the water table to 20 m above the top of the model. Find the hydrostatic pressure distribution and the displacement of the cuboid at the top; the displacement analytical solution is zero, and the analytical solution for the hydrostatic pressure distribution is given by

$$p^{(3)} = p_b^{(3)} \left(1 - \frac{y}{H}\right) + p_t^{(3)} \frac{y}{H} \quad (44)$$

where $p_b^{(3)}$ is the fluid pressure at the bottom of the model at stage (3), and $p_t^{(3)}$ is the fluid pressure at the top of the model.

The calculated value from the coupled hydro-mechanical model is $-2.0\text{e}-5 \text{ m}$, which matches the analytical value (0 m) at this stage. The hydrostatic pressure distribution is shown in Fig. 13(c).

In stage (4), continue from the previous stage, reset the displacements to zero, and then apply a pore pressure of 0.5 MPa at the base. Find the displacement of the model at the top under seepage forces in the upward direction, the analytical solution of which is given by

$$u = \frac{(p_b^{(4)} - p_t^{(3)})(1 + \nu)(1 - 2\nu)}{2E(1 - \nu)} H \quad (45)$$

where $p_b^{(4)}$ is the fluid pressure at the bottom of the model in stage (4).

The calculated value from the coupled hydro-mechanical model is $+6.967\text{e}-3 \text{ m}$, which is close to the analytical value ($+7.143\text{e}-3 \text{ m}$) at this stage, with a relative error of 2.4%. The displacement distribution is shown in Fig. 13(d).

For stage (5), continue from stage (4), reset the displacements to zero, and then apply a pore pressure of 0.1 MPa at the base. Find the displacement of the cuboid at the top under seepage forces in the downward direction, the analytical solution of which is given by

$$u = \frac{(p_b^{(5)} - p_t^{(3)})(1 + \nu)(1 - 2\nu)}{2E(1 - \nu)} H \quad (46)$$

where $p_b^{(5)}$ is the fluid pressure at the bottom of the model in stage (5).

The calculated value from the coupled hydro-mechanical model is $-6.942\text{e}-3 \text{ m}$, which is close to the analytical value ($-7.143\text{e}-3 \text{ m}$) at this stage, with a relative error of 2.8%. The displacement distribution is shown in Fig. 13(e).

8.3. Hydraulic fracturing of a disc with one injection hole

8.3.1. Rock sample model and loading conditions

Fig. 14 shows a disc with a radius of 5 m and a thickness of 0.3 m. There is an injection hole with a radius of 0.5 m at the center of the disc. The fluid pressure at the injection hole is fixed at

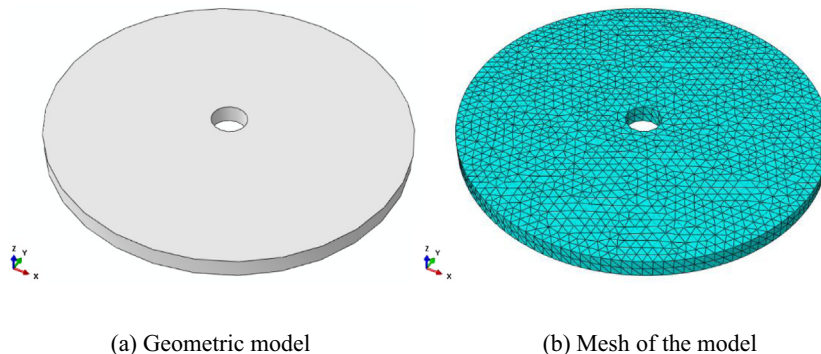


Fig. 14. Geometry and mesh of a disc with one injection hole.

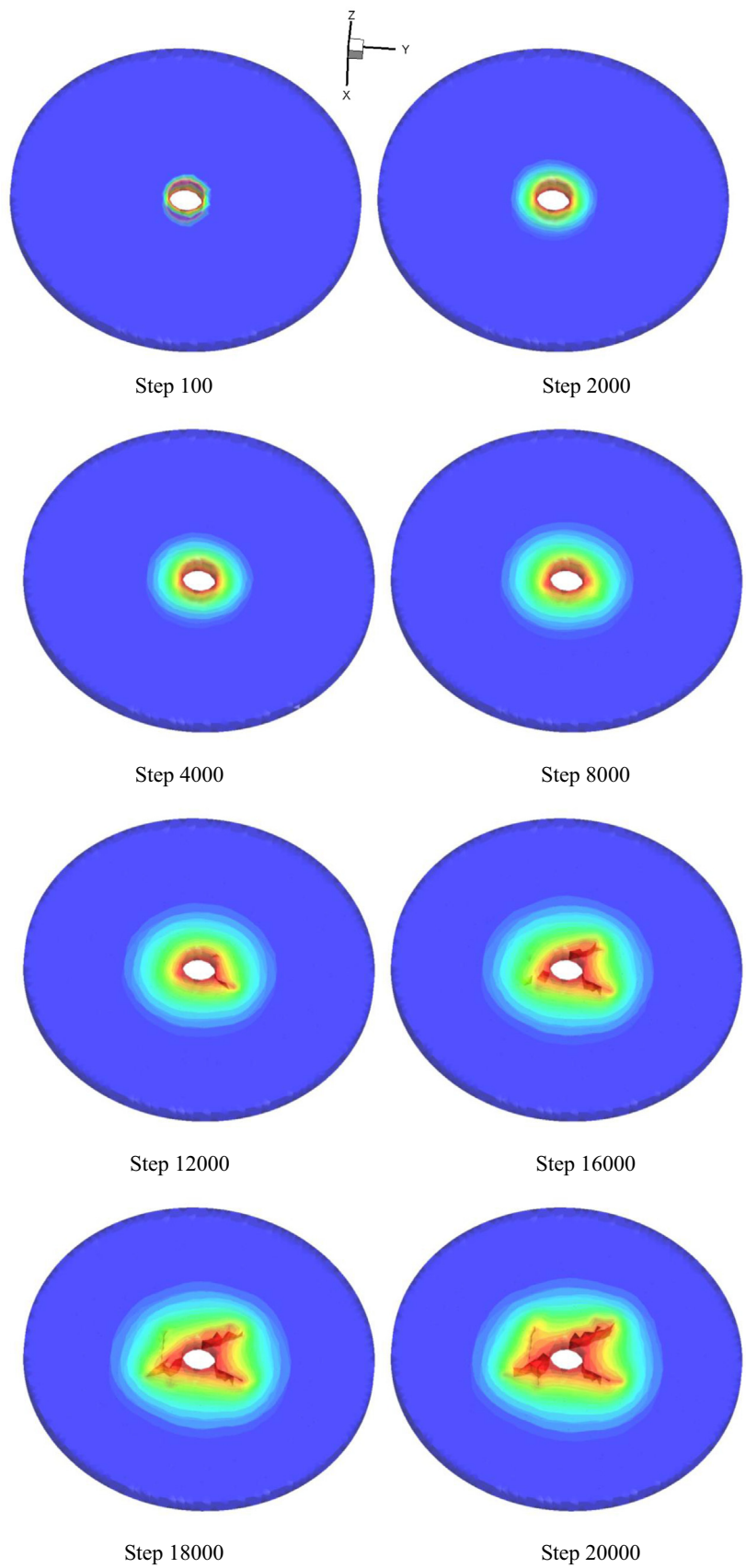


Fig. 15. Evolution of fluid pressure distribution and crack propagation with time steps during hydraulic fracturing.

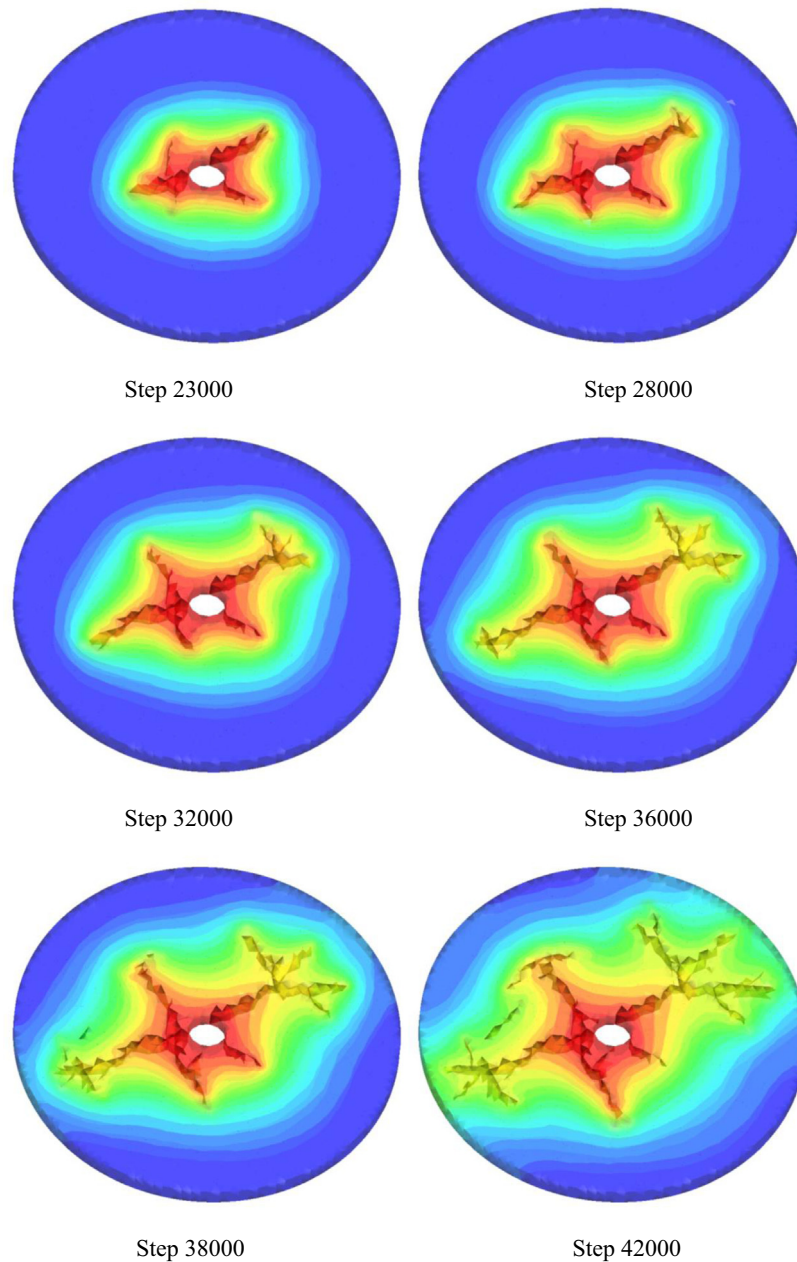


Fig. 15 (continued)

3.0 MPa, while the outer boundary surface of the disc is free. There is no other external load on the disc. The bottom and top boundary surfaces are impermeable. The entire model was meshed into 19,966 tetrahedral elements and 42,865 joint elements with a size of 0.25 m.

8.3.2. Calibration of the microscopic input parameters

The microscopic input parameters for the simulation can be calibrated using the Brazilian test (BD) and the uniaxial compression test (UCS) [53]. The microscopic input parameters used in this paper include the rock density $\rho = 2600 \text{ kg/m}^3$, the elastic modulus $E = 55 \text{ GPa}$, Poisson's ratio $\nu = 0.25$, the tensile strength of the joint element $f_t = 6 \text{ MPa}$, the cohesion of the joint element $c = 20 \text{ MPa}$, the internal friction angle $\phi = 30^\circ$, the Mode I fracture energy release rate $G_{fI} = 2.0 \text{ J/m}^2$, the Mode II fracture energy release rate $G_{fII} = 10 \text{ J/m}^2$, the normal penalty parameter $p_n = 55 \text{ GPa}$, the tangential penalty parameter $p_t = 5.5 \text{ GPa/m}$, the bulk modulus of

the fluid $K_w = 2.2 \text{ GPa}$, the viscosity coefficient of the fluid $\mu_f = 1 \text{ mPa s}$, and the fluid density $\rho_f = 1000 \text{ kg/m}^3$.

Here, we describe how to obtain the microscopic parameter associated with the seepage calculation. In Section 5, it was noted that the macroscopic permeability of the numerical rock model cannot be directly determined; instead, the initial hydraulic aperture a_0 of the joint elements is calibrated by the permeability test simulation of the numerical rock model in Fig. 8. We use the calibration method (with the mesh size of the model in Fig. 8 consistent with that of the model used in Fig. 14) to determine the initial hydraulic aperture a_0 , which makes the macroscopic permeability of the numerical rock model consistent with the permeability of the actual rock sample. It is known that the macroscopic permeability of the actual rock sample is $1.887 \times 10^{-13} \text{ m}^2$. Through the simulation of the permeability test, the permeability of the rock model is exactly the same as the permeability of the actual rock sample if $a_0 = 5 \times 10^{-5} \text{ m}$.

8.3.3. Simulation results

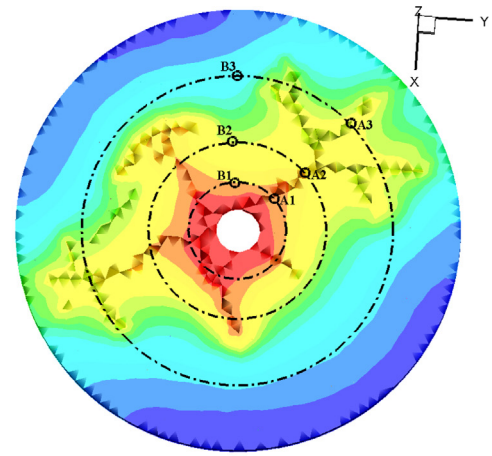
As shown in Fig. 15, although no cracks are produced within the rock sample at the beginning, the rock surrounding the injection hole still has a fluid pressure distribution due to the permeability of the rock matrix (see step 2000 in Fig. 15). With the increasing of the time steps, fluid infiltrates into the rock around the injection hole, and the fluid pressure exhibits a circular distribution. The range of the fluid pressure distribution increases gradually, as shown from step 100 to step 8000 in Fig. 15. As observed from the above results, the proposed model can consider the permeability of the rock matrix. As the number of time steps increases, cracks began to form in the rock around the injection hole, as shown in step 12,000 in Fig. 15. With the emergence of cracks, the distribution of the fluid pressure begins to change to a narrow distribution along the newly generated cracks. The fluid pressure in the rock around the cracks significantly increases, as shown from step 26,000 to step 42,000 in Fig. 15. This is because the newly generated cracks (i.e., joint elements broken) make the hydraulic aperture of the broken joint elements significantly increase. According to the cubic law, the permeability of the joint elements significantly increases compared with the intact rock (corresponding to the zone with unbroken joint elements), and the fluid flows more easily along the broken joint element (i.e., crack) than along the unbroken joint element (corresponding to the rock matrix). It can be observed that the proposed FDEM-flow3D model in this paper takes into account the permeability of both the rock matrix and fractures.

To further analyze the variation of the fluid pressure, we take some monitoring points on circumferences with radii of 1.1 m, 2 m and 3.5 m, where points A1–A3 are located on the boundary of the newly generated cracks and B1–B3 are on the intact rock blocks all the time; points A1 and B1 are located on a circumference with a radius of 1.1 m; points A2 and B2 are located on a circumference with a radius of 2 m, and points A3 and B3 are located on a circumference with a radius of 3.5 m, as shown in Fig. 16(a). We observe the variation of the fluid pressure at those monitoring points during hydraulic fracturing.

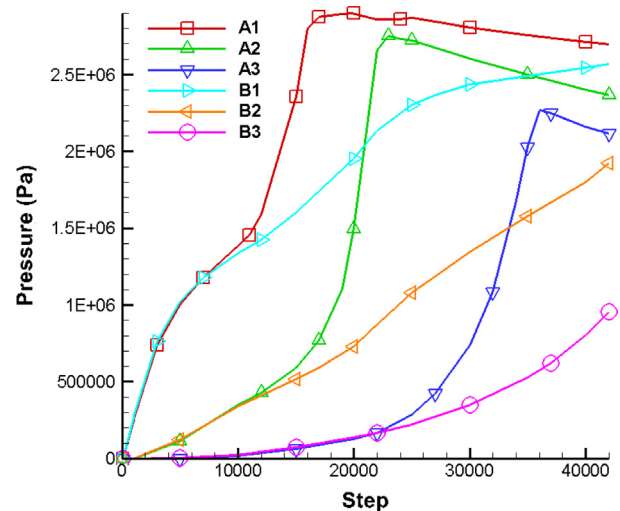
As shown in Fig. 16(b), from monitoring points A1, A2, and A3, we can find that the closer a point is to the injection hole, the larger the fluid pressure is at the same moment. There exists a sharply increasing phase in the fluid pressure curve with the time steps when the cracks extend to the monitoring points (i.e., the joint elements at the monitoring points break). Due to the cohesive effect loss of the joint elements, the hydraulic aperture of the broken joint elements increase under fluid pressure. Thus, the permeability of those joint elements increase, and more fluid enters into the monitoring points along the cracks, which results in a sharp increase in the pressure of the monitoring points. The fluid pressures at points A1, A2 and A3 reach peaks at steps 20,000, 23,000 and 26,000, respectively, and then gradually decreases because these cracks continue to extend and make the net inflows into the three monitoring points decrease.

Similarly, for points B1, B2, and B3, the larger the distance of the monitoring points from the injection hole, the larger the pressure is. However, the fluid pressure at the three monitoring points of group B gradually increases, and no sharp increase appears.

Next, we compare the monitoring points with same distance from the center of the injection hole in groups A and B. Take points A2 and B2 as an example. The fluid pressure of these points A2 and B2 is not equal to zero at step 8000 or the previous time step, although there are no cracks generated in the disc, as shown in steps 1000–8000 in Fig. 15. This means that the fluid has infiltrated into the rock around the injection hole because of the rock matrix permeability. Moreover, it can be found that the fluid pressures at monitoring points A2 and B2 are almost the same before the cracks extend to monitoring point A2 (at step 12,000). This is because the distances of points A2 and B2 from the center of the injection hole



(a) Locations of monitoring points (Plan view).



(b) Variation of fluid pressure

Fig. 16. Variation of fluid pressure at monitoring points with time steps.

are equal and the initial permeability of the disc is the same, so the fluid pressure displays a circular distribution, i.e., the pressures at points A2 and B2 are almost the same. When the cracks extended to A2, the joint elements at point A2 break. Then, the hydraulic aperture of the joint elements significantly increases, leading to a permeability enhancement, so that fluid easily flows into monitoring point A2 along the crack. Thus, the fluid pressure at point A2 increases considerably. This is the reason why there is a sharply increasing phase at the fluid pressure curve of point A2. However, because point B2 is located on the rock matrix all the time and the permeability of the rock matrix is relatively small, the fluid cannot easily infiltrate into point B2, so that the fluid pressure at this point is still small. Therefore, the fluid pressure curves of B2 do not have the sharply increasing phase like that of the points in group A. Points A1 and B1 and A3 and B3 also show a similar feature. By this example, we demonstrated that the FDEM-flow3D model in this study can simultaneously consider the pore seepage of the rock matrix and the fracture seepage.

9. Conclusions

- (1) An FDEM-flow3D model that can consider the pore seepage of a rock matrix and the fracture seepage is proposed to sim-

ulate three-dimensional hydraulic fracturing. The model can consider the permeability of the rock matrix and fractures using a simple pure fracture seepage.

- (2) A calibration method that determines the initial hydraulic aperture of the joint elements in a numerical model to characterize the permeability of a rock matrix is proposed. The correctness of the seepage algorithm and the calibration method are verified using a unidirectional flow numerical example with an analytical solution. A simple poroelastic example with an analytical solution is also provided to verify the hydro-mechanical coupled model.
- (3) Finally, we simulated a hydraulic fracturing example; the results show that the fluid pressure also distributes in the rock around the injection hole before the crack generation. When cracks are generated, the fluid pressures of the monitoring points at the cracks significantly increase, whereas the fluid pressures of points with the same radial distance in intact rock increase only a little. Using this example, we demonstrate that FDEM-flow3D can consider the permeability of the rock matrix and the fracture at the same time.

The proposed FDEM-flow3D model can succinctly solve a complex problem such as hydraulic fracturing. The fluid pressure distribution and the crack initiation and propagation during hydraulic fracturing can be well reproduced by this model. In the future, this model will be applied to solve the problem of fluid leak-off.

Acknowledgements

This work has been supported by the China Postdoctoral Science Foundation under grant number 2015M580953; the National Natural Science Foundation of China under grant numbers 11602006 and 11572009; and the Natural Basic Research Program of China under grant numbers 2011CB013505 & 2014CB047100.

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